

Pair	Source (X)	Target (Y)	Hypothesis	CTE (X→Y)	CTE (Y→X)	p	Lag
P1	PTEN-T1	PIP3	PTEN catalyzes PIP3 degradation	0.100	0.024	1.000	2
P2	PIP3	PTEN-T1	PIP3 inhibits PTEN recruitment	0.024	0.100	1.000	2
P3	PI3K	PIP3	PI3K produces PIP3	0.058	0.000	0.274	2
P4	PTEN-T1	PTEN-T2	PTEN conformational switching	0.000	0.183	1.000	5
P5	PIP3	N(complexes)	PIP3 drives biosensor binding	0.008	0.113	0.378	2
P6	N(complexes)	$\bar{\tau}$	Binding causes lifetime change	0.141	0.018	0.005***	1
P7	MSD	N(complexes)	Diffusion limits binding encounters	0.034	0.007	0.910	2
P8	N(bleached)	Intensity	Bleaching causes signal loss	0.043	0.002	0.403	5
P9	$\bar{E}$ (FRET)	N(bleached)	FRET modulates bleaching rate	0.000	0.058	0.851	2
P10	MSD	$\bar{\tau}$	Mobility causes lifetime (negative control)	0.069	0.002	0.254	2